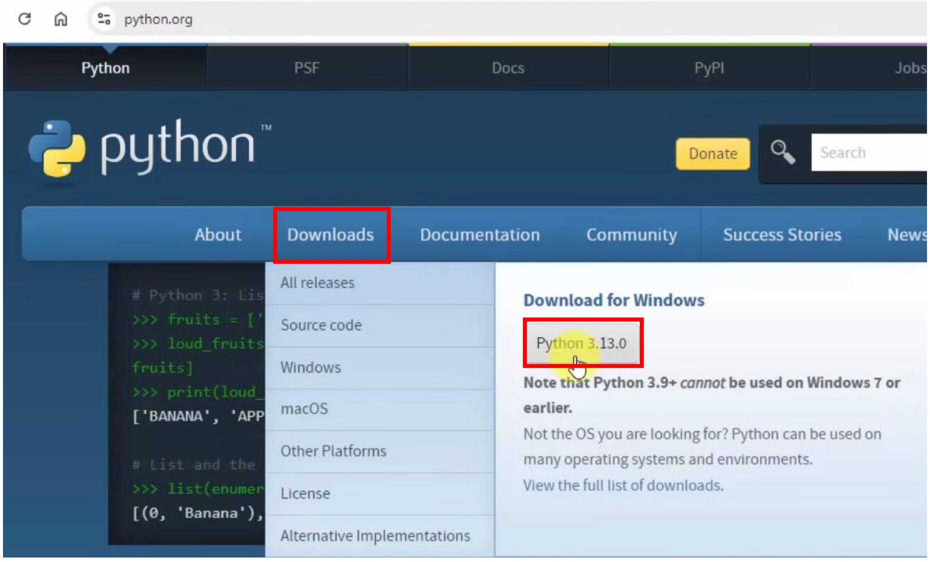
Install Python environment on your Windows

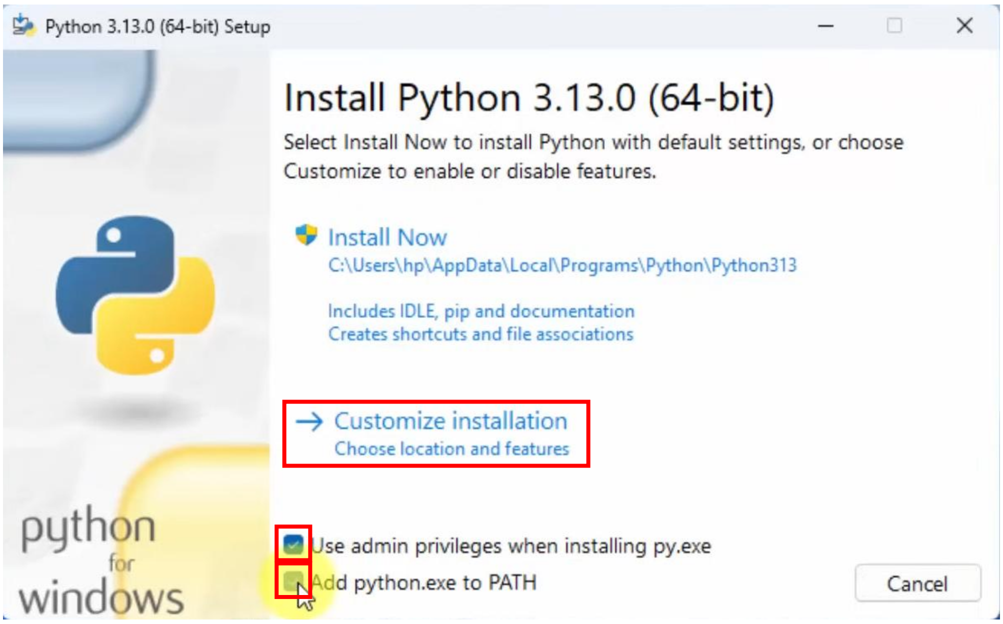
1. Click Downloads > Python 3.x.x



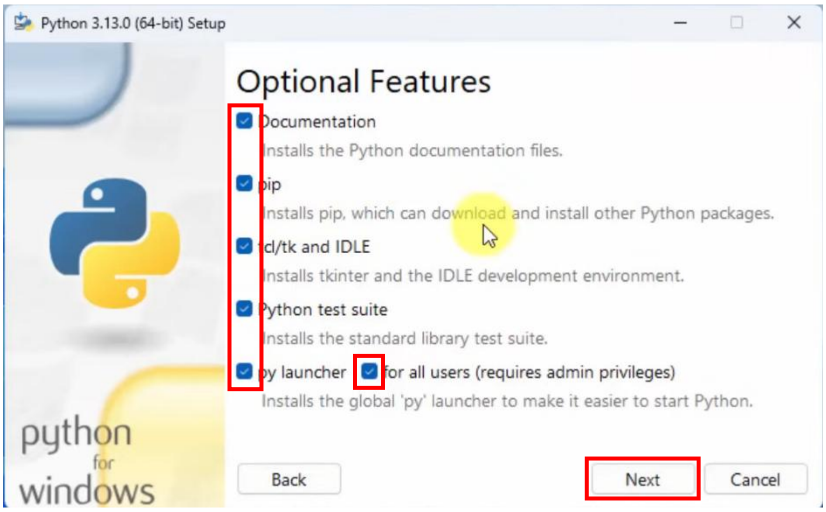
2. Check:

- Use admin privileges when installing py.exe
- Add python.exe to PATH

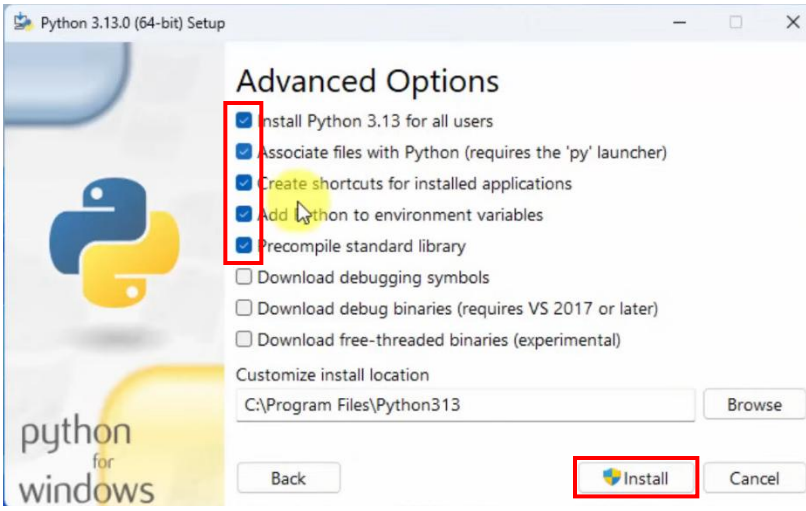
> Customize installation



3. Check all > Next

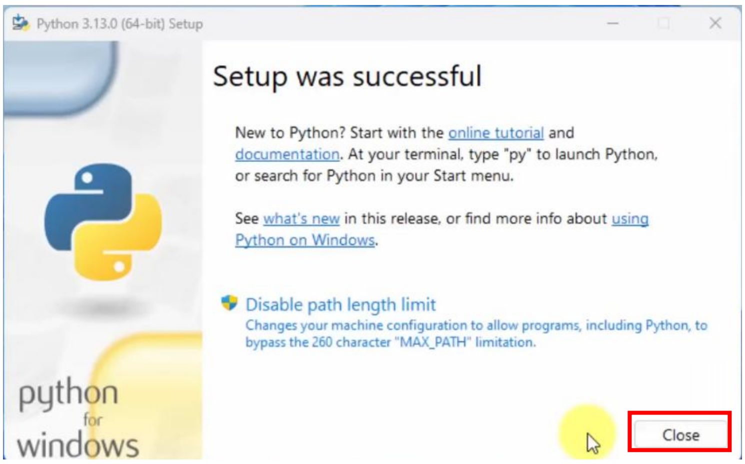


4. Advanced options



Check:
- Install Python 3.13 for all users
- Associate files with Python
- Create shortcuts for installed applications
- Add Python to environment variables
- Precompile standard library
> Install

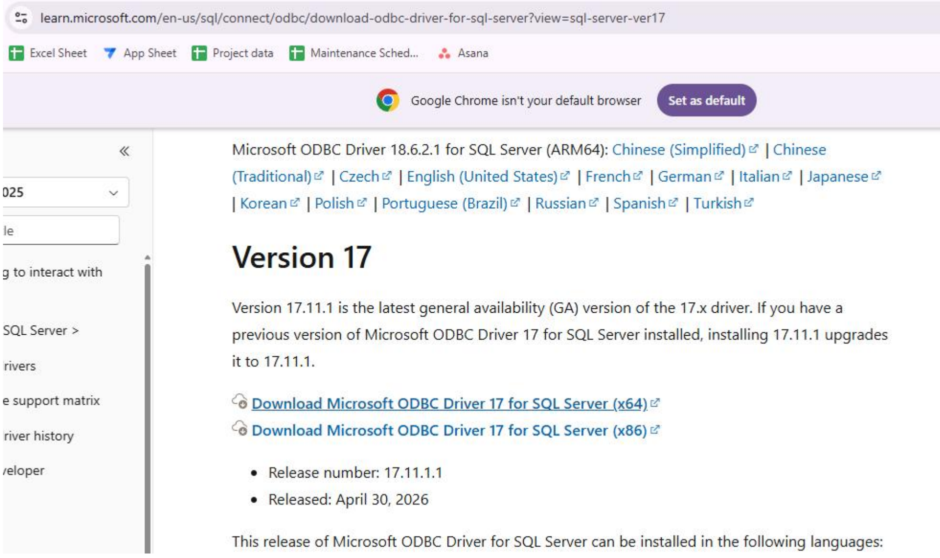
5. **Complete installation: click Close**



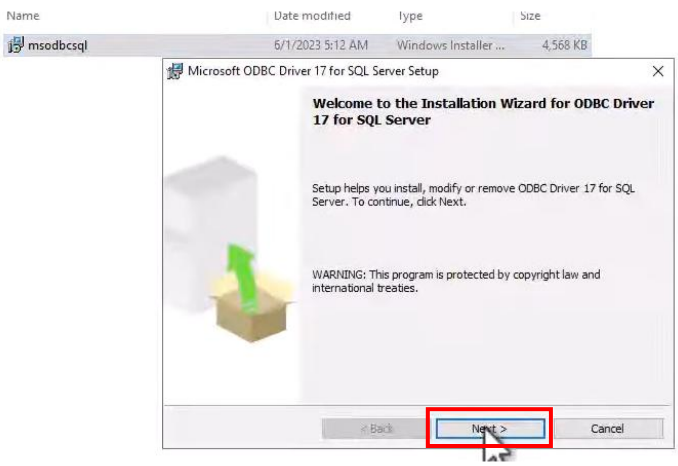
ODBC SQL Server Connection

6. Install ODBC Driver17

Url: https://learn.microsoft.com/en-us/sql/connect/odbc/download-odbc-driver-for-sql-server?view=sql-server-ver17

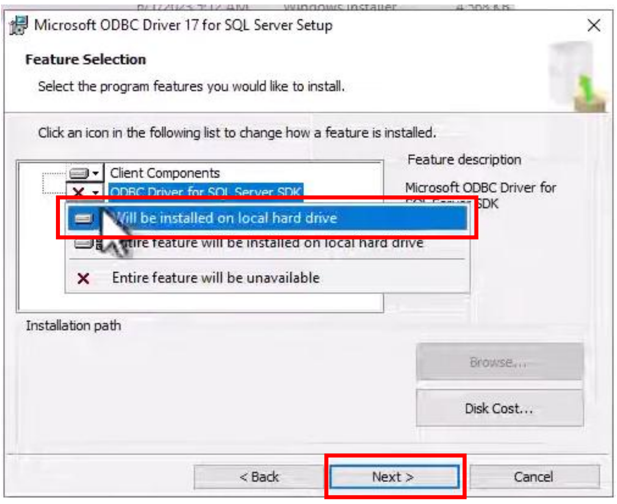


Click next



Check: I accept the term in the license agreement > Next

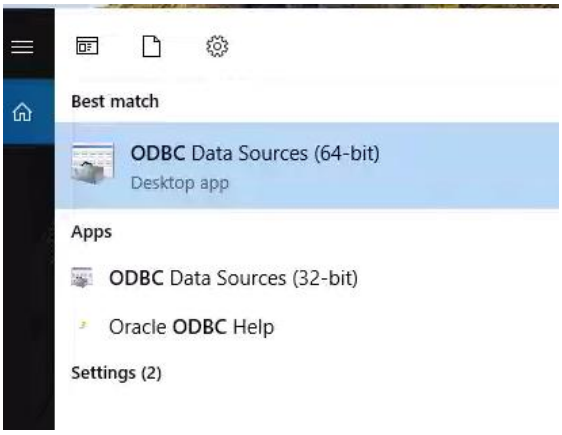
Click ODBC Driver for SQL Server SDK > Will be installed on local hard drive > Next



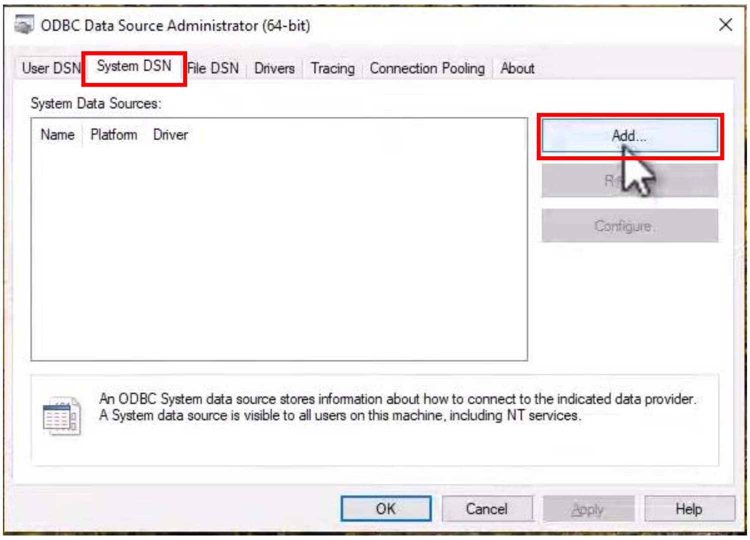
Next > Finish

Configure ODBC

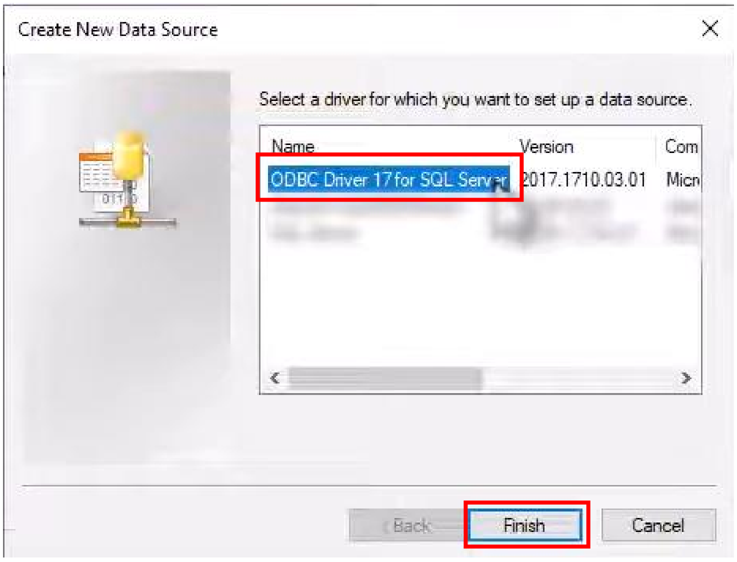
7. Search ODBC Data Sources (64-bit)



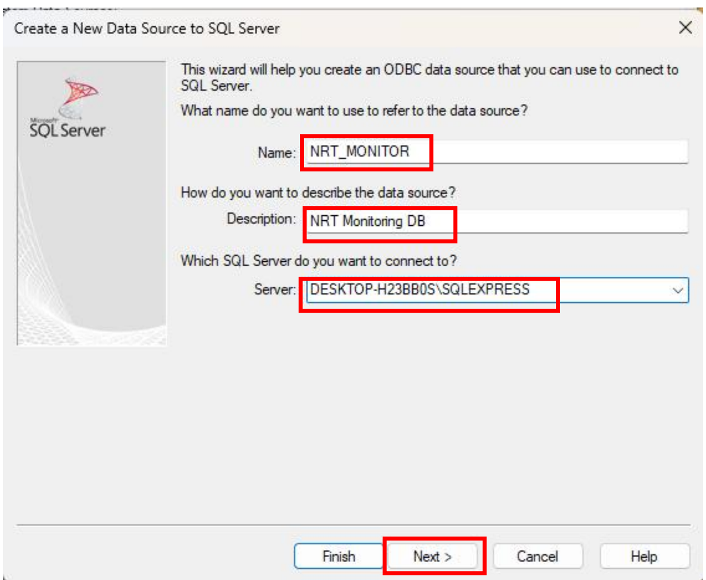
Click System DSN > Add



Choose ODBC Driver 17 for SQL Server

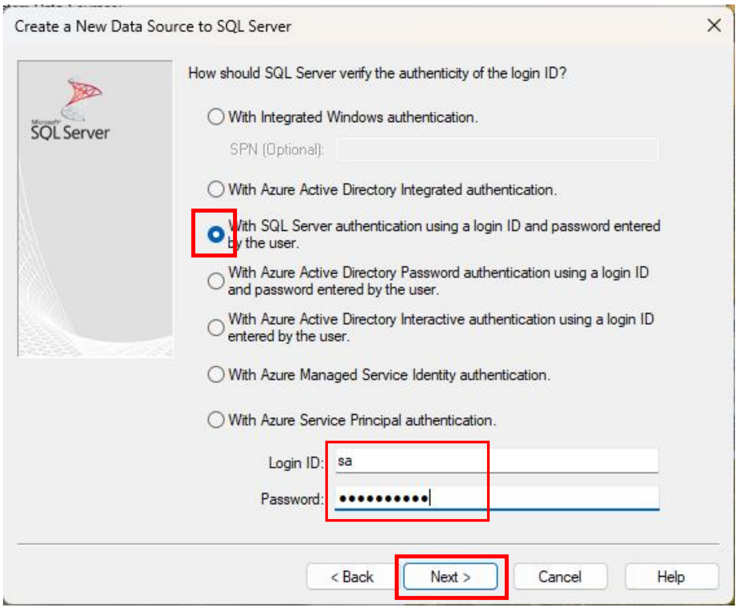


Fill in the details accordingly



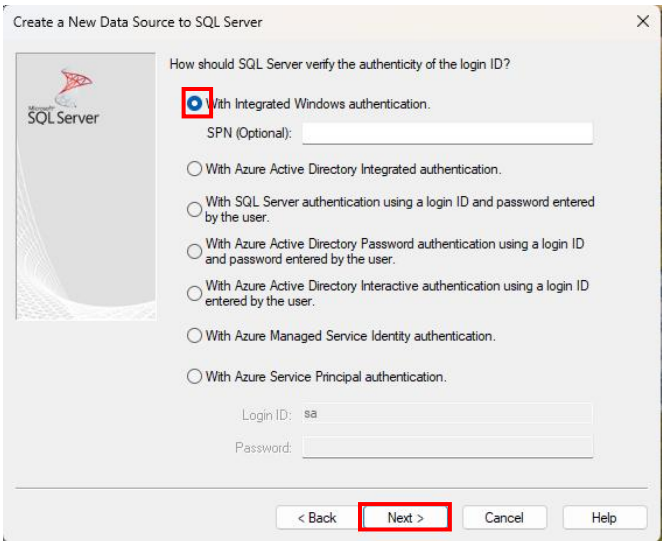
Name: NRT_MONITOR
Description: NRT Monitoring DB
Server: <Choose your ssms server name>
> Next

If your server is authenticated checked: “With SQL Server authentication using a login ID and password entered by the user” Enter your login ID and password > Next

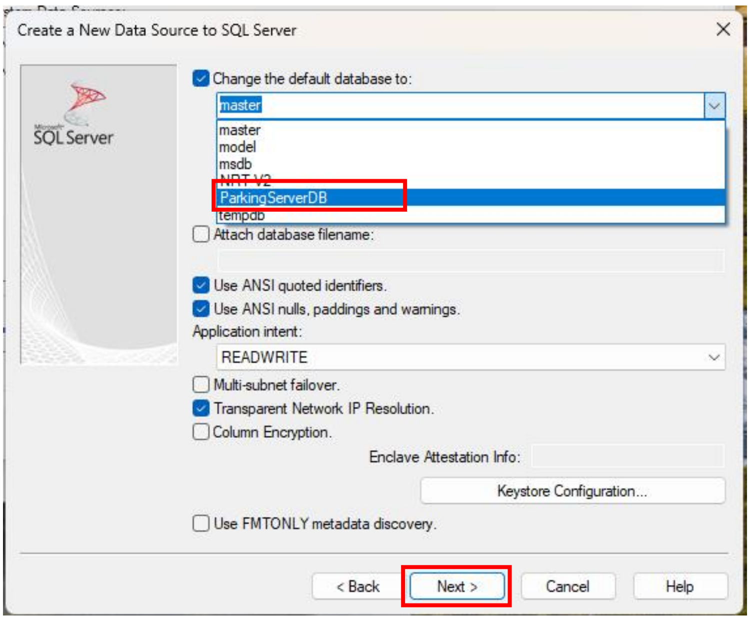


Login ID: sa
Password: qwerty7890

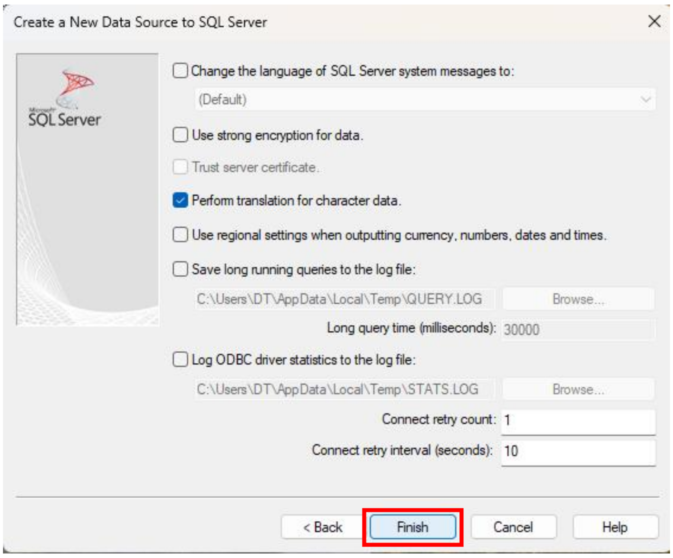
If your server is unauthenticated, choose “With Integrated Windows authentication” > Next



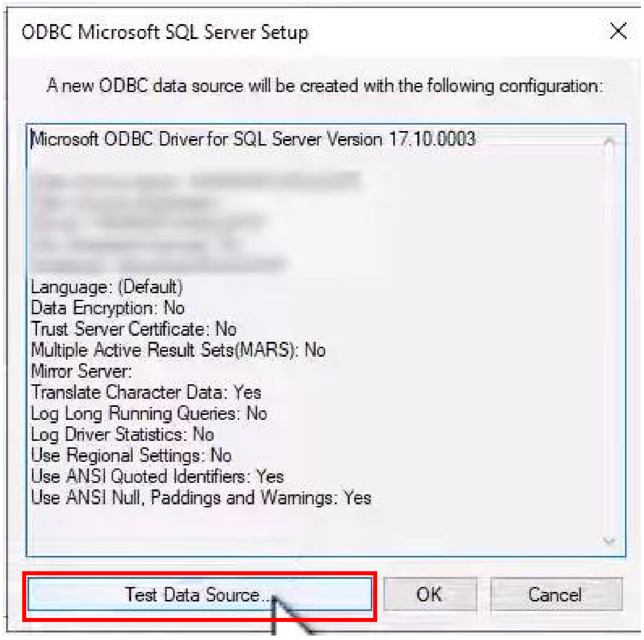
Change the default database to: ParkingServerDB > Next



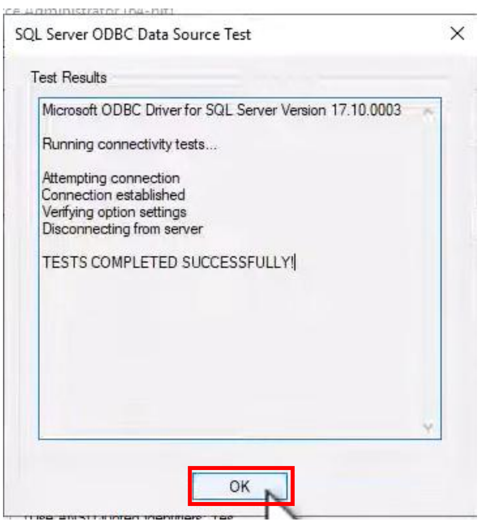
Click Finish



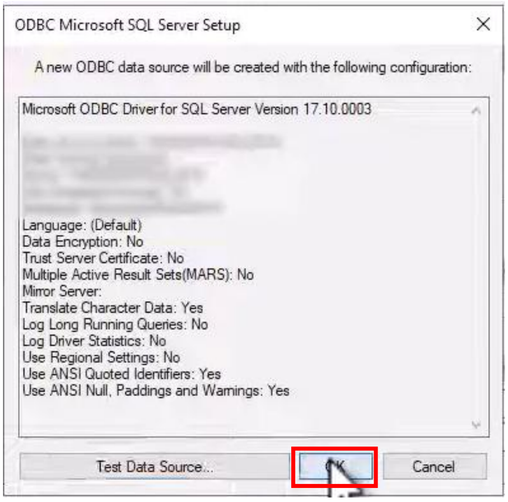
Click Test Data Source



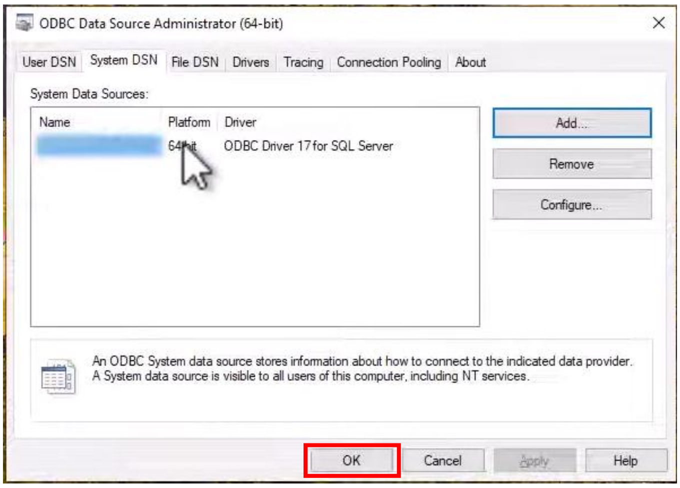
If you see ‘Connection established’, it means the connection is successful > click OK



Click OK



Click OK



Run Instruction

1. **Change directory**

```
cd NRT_Monitor
```

2. **Create a virtual environment**

```
py -m venv venv
.\venv\Scripts\activate  # On Linux Mac OS: source venv/bin/activate
```

- If errors occurred, run this:

```
Set-ExecutionPolicy -ExecutionPolicy RemoteSigned -Scope CurrentUser
```

3. **Install dependencies**

```
pip install -r requirements.txt
```

4. Credentials configuration
In the Agent.py

```
SQL_SERVER = r"(local)\SQLEXPRESS" # eg.,: “(local)\SQLEXPRESS” or “DESKTOP-TAICUPD\SQLEXPRESS”
DATABASE = "NRT-V2"
SITE_NAME = "eg.,: Penang Courtyard"
SITE_CODE = DATABASE
API_URL = "https://script.google.com/macros/s/AKfycbxIAwFbkot1Q5X54EbhLik3NLq85KcpYVkeOyh-9Rjgui0nrcI5zIvQaVckHQWAxZju/exec"
API_KEY = "nrt_8F2xQ9mL7vP3zK1cR6wT4yH0bN5sJ8"
DASHBOARD_API_URL = "http://127.0.0.1:5000/api/nrt-status"

ALERT_THRESHOLD_HOURS = 48
COLLECTION_INTERVAL_MINUTES = 40
STARTUP_LAUNCHER_NAME = "NRT Monitor Collector.cmd"

CONNECTION_STRING = (
    "DRIVER={ODBC Driver 17 for SQL Server};"
    f"SERVER={SQL_SERVER};"
    f"DATABASE={DATABASE};"
    "UID=sa;"
    "PWD=qwerty7890;"
    "Trusted_Connection=yes;"
)
```

5. **Run the app**

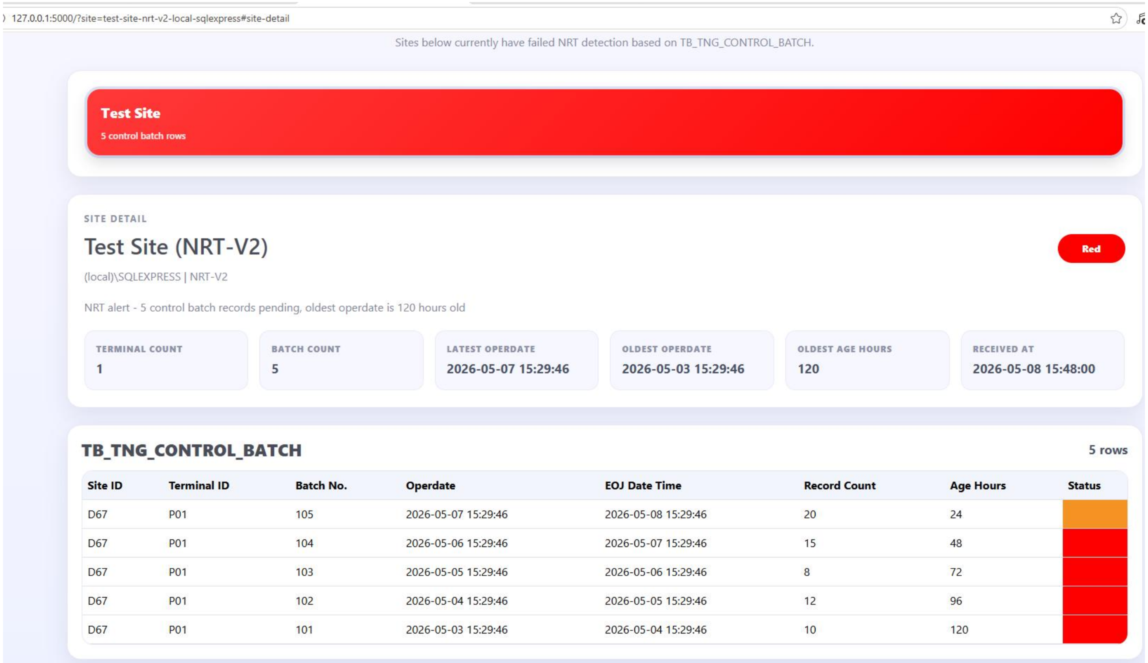
Place both program one in the Central Server, another in the client site
On the Central Server, run:

```
python app.py
```

On the client side, run:

```
python agent.py
```

Output:



127.0.0.1:5000/?site=test-site-nrt-v2-local-sqlexpress#site-detail

Sites below currently have failed NRT detection based on TB_TNG_CONTROL_BATCH.

Test Site

5 control batch rows

SITE DETAIL

Test Site (NRT-V2)

Red

(local)\SQLEXPRESS | NRT-V2

NRT alert - 5 control batch records pending, oldest operdate is 120 hours old

TERMINAL COUNT	BATCH COUNT	LATEST OPERDATE	OLDEST OPERDATE	OLDEST AGE HOURS	RECEIVED AT
1	5	2026-05-07 15:29:46	2026-05-03 15:29:46	120	2026-05-08 15:48:00

TB_TNG_CONTROL_BATCH

5 rows

Site ID	Terminal ID	Batch No.	Operdate	EOJ Date Time	Record Count	Age Hours	Status
D67	P01	105	2026-05-07 15:29:46	2026-05-08 15:29:46	20	24	
D67	P01	104	2026-05-06 15:29:46	2026-05-07 15:29:46	15	48	
D67	P01	103	2026-05-05 15:29:46	2026-05-06 15:29:46	8	72	
D67	P01	102	2026-05-04 15:29:46	2026-05-05 15:29:46	12	96	
D67	P01	101	2026-05-03 15:29:46	2026-05-04 15:29:46	10	120	